

Problem A: Hello, ICPC!

Time limit: 1 second

Alice is the new Important Code Performance Caretaker (ICPC) at her company. Her job is to check the code written by the developers for bad performance and improve it if possible. The code is often strange since a company-wide policy is to write every piece of code in four programming languages. This is to ensure that the code runs on as many platforms as possible although Alice doubts that this is the best way to do it. Either way, she found the following code written in C++, Python, Java, and Haskell and wants to improve it. Please help her rewrite the code in one of the languages such that it prints the same result for all inputs but always runs within a second.

C++:

```
#include <algorithm>
#include <iostream>
#include <vector>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<long long> a(n);
    for (int i = 0; i < n; i++) cin >> a[i];

    vector<long long> p;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            for (int k = 0; k < n; k++) {
                p.push_back(a[i]*a[j]*a[k]);
            }
        }
    }

    sort(p.begin(), p.end());

    cout << p[p.size()-1] << endl;
    return 0;
}
```

Python 3:

```
n = int(input())
a = list(map(int, input().split()))

p = []
for i in range(n):
    for j in range(n):
        for k in range(n):
            p.append(a[i]*a[j]*a[k])
p.sort()

print(p[-1])
```

Java:

```
import java.util.*;

public class HelloGCPC {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);

        int n = s.nextInt();

        long[] a = new long[n];
        for(int i = 0; i < n; i++) a[i] = s.nextLong();

        LinkedList<Long> p = new LinkedList<>();
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                for (int k = 0; k < n; k++) {
                    p.add(a[i]*a[j]*a[k]);
                }
            }
        }

        Collections.sort(p);

        System.out.println(p.get(p.size()-1));
    }
}
```

Haskell:

```
import qualified Data.ByteString.Char8 as C
import Data.List

readInt :: C.ByteString -> Int
readInt s = case C.readInt s of Just (x,_) -> x

main :: IO ()
main = C.getContents >>= print . solve . map readInt . C.words

solve :: [Int] -> Int
solve (n:as) = last $ sort [ x*y*z | x <- as, y <- as, z <- as ]
```

Input

The input consists of:

- One line with an integer n ($1 \leq n \leq 10^5$), the number of integers to be read.
- One line with n integers $a_1, \dots, a_n$ ($0 \leq a_i \leq 10^6$ for each i).

Output

Output the same number as the four programs above.

Sample Input 1

```
3
1 2 3
```

Sample Output 1

```
27
```

Sample Input 2

```
5
344 823 3950 167 4784
```

Sample Output 2

```
109489762304
```